

# DATA SHEET

**SAA7152**

**Digital Video Comb Filter (DCF)**

Product specification  
File under Integrated Circuits, IC02

August 1996

## Digital Video Comb Filter (DCF)

## SAA7152



## FEATURES

- Comb filter circuit for luminance and chrominance separation
- Applicable for standards
  - PAL B/G, M and N
  - PAL 4.43 (525 lines, 60 Hz)
  - NTSC M and N
  - NTSC 4.43 (50 and 60 Hz)
- Luminance and chrominance bypasses with short delay in case of no filtering
- Line-locked system clock; CCIR-compatible
- I<sup>2</sup>C-bus controlled

## GENERAL DESCRIPTION

The CMOS digital comb filter circuit is located between video analog-to-digital converters and the video multistandard decoder SAA7151B (not applicable for SAA7191B). The two-dimensional filtering is only appropriate for standard signals from a source with constant phase relationship between subcarrier signal and horizontal frequency. The comb-filter has to be switched off for VTR-signals and for separate VBS and C signals. In VCR and S-Video operation the luminance low-pass and the chrominance bandpass parts can still be used for noise reduction purposes. The processing delay is:

21 × LL27 clocks in active mode, or

3 × LL27 in short delay bypass mode (BYPSS = 1)

## QUICK REFERENCE DATA

| SYMBOL           | PARAMETER                           | MIN.           | TYP. | MAX. | UNIT |
|------------------|-------------------------------------|----------------|------|------|------|
| V <sub>DD</sub>  | supply voltage (pins 11, 34, 44)    | 4.5            | 5.0  | 5.5  | V    |
| I <sub>P</sub>   | total supply current                | –              | 85   | 180  | mA   |
| V <sub>i</sub>   | input levels                        | TTL-compatible |      |      |      |
| V <sub>o</sub>   | output levels                       | TTL-compatible |      |      |      |
| LL27             | typical system clock frequency      | –              | 27   | –    | MHz  |
| T <sub>amb</sub> | operating ambient temperature range | 0              | –    | 70   | °C   |

## ORDERING INFORMATION

| EXTENDED<br>TYPE NUMBER | PACKAGE |              |          |                       |
|-------------------------|---------|--------------|----------|-----------------------|
|                         | PINS    | PIN POSITION | MATERIAL | CODE                  |
| SAA7152                 | 44      | PLCC         | plastic  | SOT187 <sup>(1)</sup> |

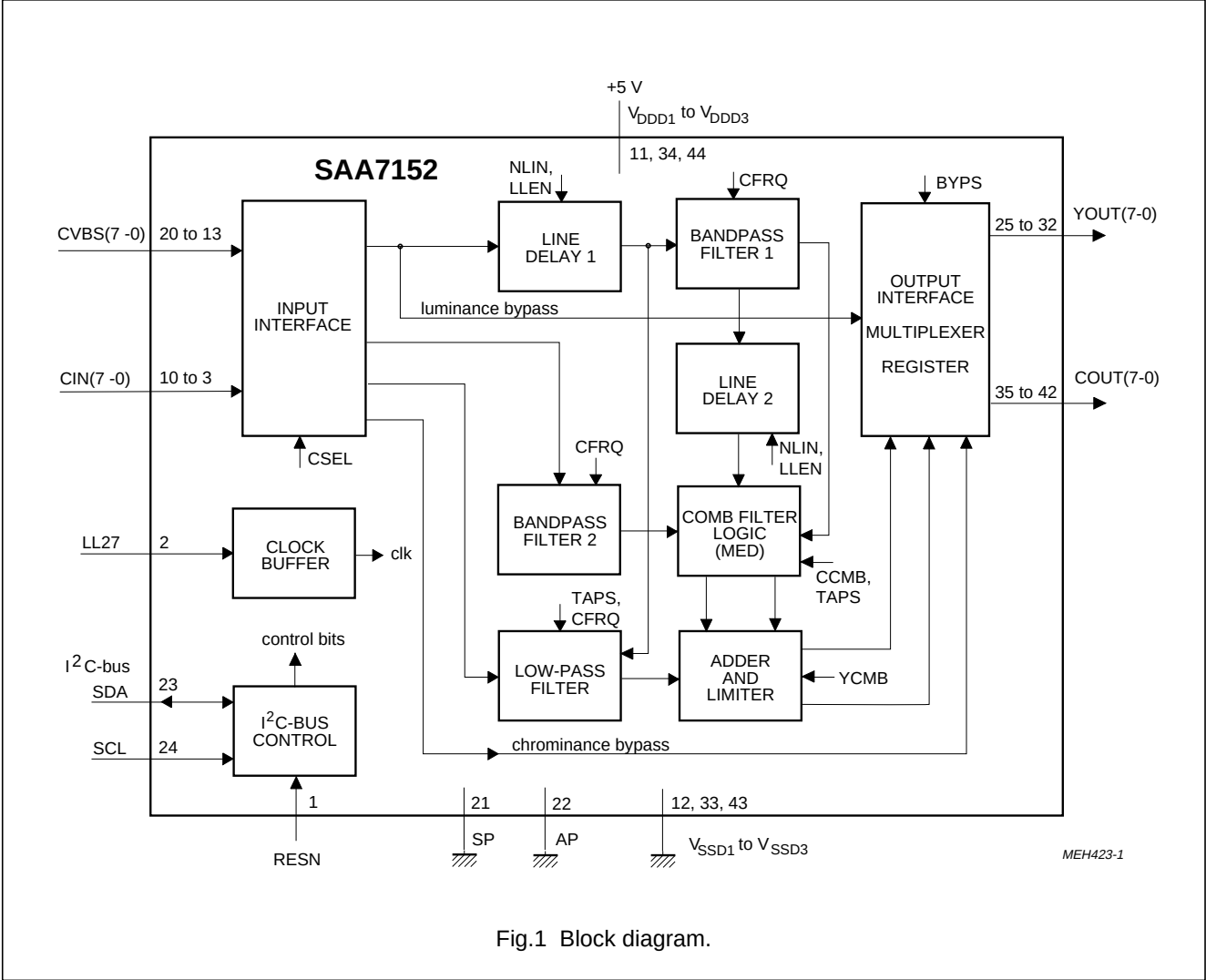
## Note

1. SOT187-2; 1997 January 06.

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BLOCK DIAGRAM



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## PINNING

| SYMBOL           | PIN | DESCRIPTION                                  |
|------------------|-----|----------------------------------------------|
| RESN             | 1   | reset input; active low                      |
| LL27             | 2   | line-locked system clock input (27 MHz)      |
| CIN0             | 3   | chrominance input data bits CIN0 to CIN7     |
| CIN1             | 4   |                                              |
| CIN2             | 5   |                                              |
| CIN3             | 6   |                                              |
| CIN4             | 7   |                                              |
| CIN5             | 8   |                                              |
| CIN6             | 9   |                                              |
| CIN7             | 10  |                                              |
| V <sub>DD1</sub> | 11  | +5 V supply input                            |
| V <sub>SS1</sub> | 12  | ground 1 (0 V)                               |
| CVBS0            | 13  | CVBS input data bits 0 to 7                  |
| CVBS1            | 14  |                                              |
| CVBS2            | 15  |                                              |
| CVBS3            | 16  |                                              |
| CVBS4            | 17  |                                              |
| CVBS5            | 18  |                                              |
| CVBS6            | 19  |                                              |
| CVBS7            | 20  |                                              |
| SP               | 21  | connected to ground (shift pin for testing)  |
| AP               | 22  | connected to ground (action pin for testing) |
| SDA              | 23  | I <sup>2</sup> C-bus data line               |
| SCL              | 24  | I <sup>2</sup> C-bus clock line              |
| YOUT7            | 25  | luminance (Y) output data bits 7 to 0        |
| YOUT6            | 26  |                                              |
| YOUT5            | 27  |                                              |
| YOUT4            | 28  |                                              |
| YOUT3            | 29  |                                              |
| YOUT2            | 30  |                                              |
| YOUT1            | 31  |                                              |
| YOUT0            | 32  |                                              |
| V <sub>SS2</sub> | 33  | ground 2 (0 V)                               |
| V <sub>DD2</sub> | 34  | +5 V supply input 2                          |

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| SYMBOL           | PIN | DESCRIPTION                             |
|------------------|-----|-----------------------------------------|
| COUT7            | 35  | chrominance (C) output data bits 7 to 0 |
| COUT6            | 36  |                                         |
| COUT5            | 37  |                                         |
| COUT4            | 38  |                                         |
| COUT3            | 39  |                                         |
| COUT2            | 40  |                                         |
| COUT1            | 41  |                                         |
| COUT0            | 42  |                                         |
| V <sub>SS3</sub> | 43  | ground 3 (0 V)                          |
| V <sub>DD3</sub> | 44  | +5 V supply input 3                     |

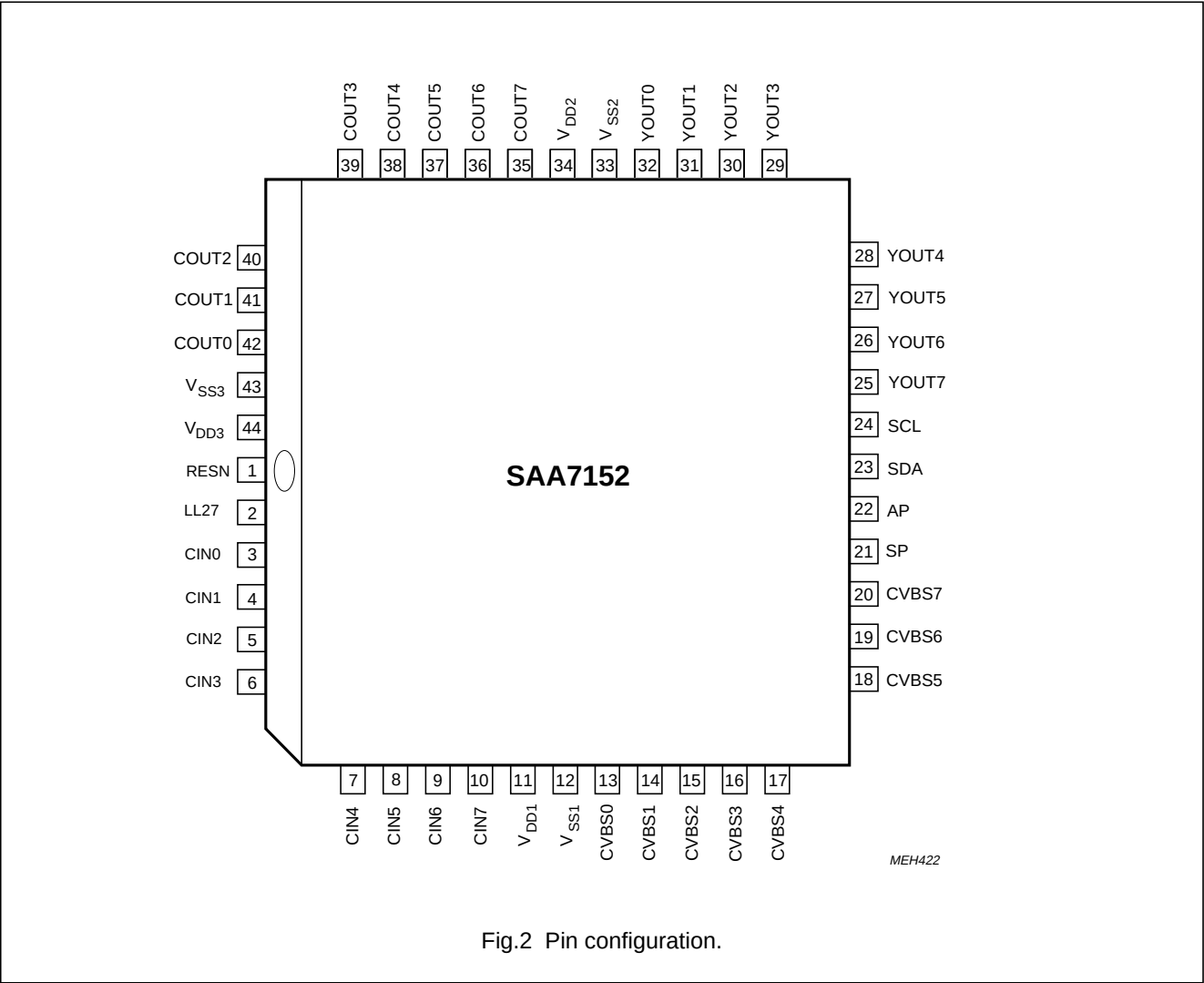
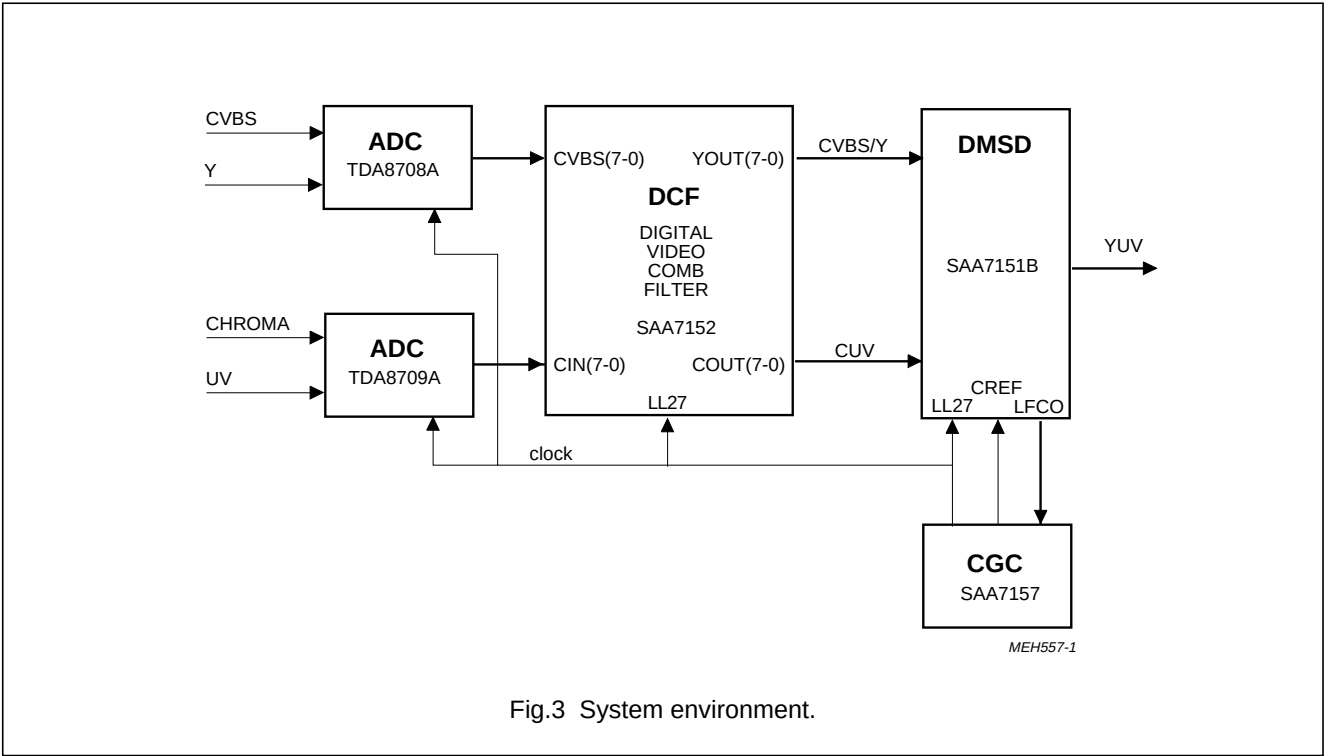


Fig.2 Pin configuration.

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I<sup>2</sup>C-BUS FORMAT

| S                         | SLAVE ADDRESS | A | SUBADDRESS                                              | A | DATA0 | A | ..... | DATAn | A | P |
|---------------------------|---------------|---|---------------------------------------------------------|---|-------|---|-------|-------|---|---|
| S                         | =             |   | start condition                                         |   |       |   |       |       |   |   |
| SLAVE ADDRESS             | =             |   | 1011 0010 (B2 h)                                        |   |       |   |       |       |   |   |
| A                         | =             |   | acknowledge, generated by the slave                     |   |       |   |       |       |   |   |
| SUBADDRESS <sup>(1)</sup> | =             |   | subaddress byte (Table 1)                               |   |       |   |       |       |   |   |
| DATA                      | =             |   | data byte (Table 1)                                     |   |       |   |       |       |   |   |
| P                         | =             |   | stop condition                                          |   |       |   |       |       |   |   |
| X                         | =             |   | read/write control bit                                  |   |       |   |       |       |   |   |
|                           |               |   | X = 0, order to write (the circuit is slave receiver)   |   |       |   |       |       |   |   |
|                           |               |   | X = 1, order to read (the circuit is slave transmitter) |   |       |   |       |       |   |   |

Note

1. If more than 1 byte DATA are transmitted, then auto-increment of the subaddress is performed.

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**Table 1** I<sup>2</sup>C-bus; subaddress and data bytes for writing (after X = 0 in address byte)

| FUNCTION | SUBADDRESS | DATA |      |      |      |      |      |      |      |
|----------|------------|------|------|------|------|------|------|------|------|
|          |            | D7   | D6   | D5   | D4   | D3   | D2   | D1   | D0   |
| Controls | 00         | BYPS | CSEL | CCMB | YCMB | TAPS | CFRQ | NLIN | LLEN |

**Function of the bits of Table 1:**

|      |                                                                     |                                                  |
|------|---------------------------------------------------------------------|--------------------------------------------------|
| BYPS | Select bypass with a short delay; all other functions are disabled: | 0 = no bypass                                    |
|      |                                                                     | 1 = comb filter bypassed (delay is 3 LLC)        |
| CSEL | Input mode select:                                                  | 0 = CVBS selected                                |
|      |                                                                     | 1 = Y/C selected                                 |
| CCMB | Select comb filtering                                               | 0 = chrominance is bandpassed                    |
|      |                                                                     | 1 = chrominance is comb-filtered                 |
| YCMB | Enable chrominance subtraction from CVBS signal:                    | 0 = disabled, CVBS/Y signal is only low-passed   |
|      |                                                                     | 1 = enabled (chrominance trap or comb filtering) |
| TAPS | Selects tap for switching Y and C to adder:                         | 0 = for bandpass/low-pass combination            |
|      |                                                                     | 1 = for comb filter active                       |
| CFRQ | Select centre frequency and matching factor of chrominance filter:  | 0 = 4.43 MHz                                     |
|      |                                                                     | 1 = 3.58 MHz                                     |
| NLIN | Select delay (number of lines):                                     | 0 = 4-line comb filter for standard PAL          |
|      |                                                                     | 1 = 2-line filter for standard NTSC              |
| LLEN | Selects the number of clocks for each line delay:                   | 0 = 1728 clocks (625 lines; 50 Hz)               |
|      |                                                                     | 1 = 1716 clocks (525 lines; 60 Hz)               |

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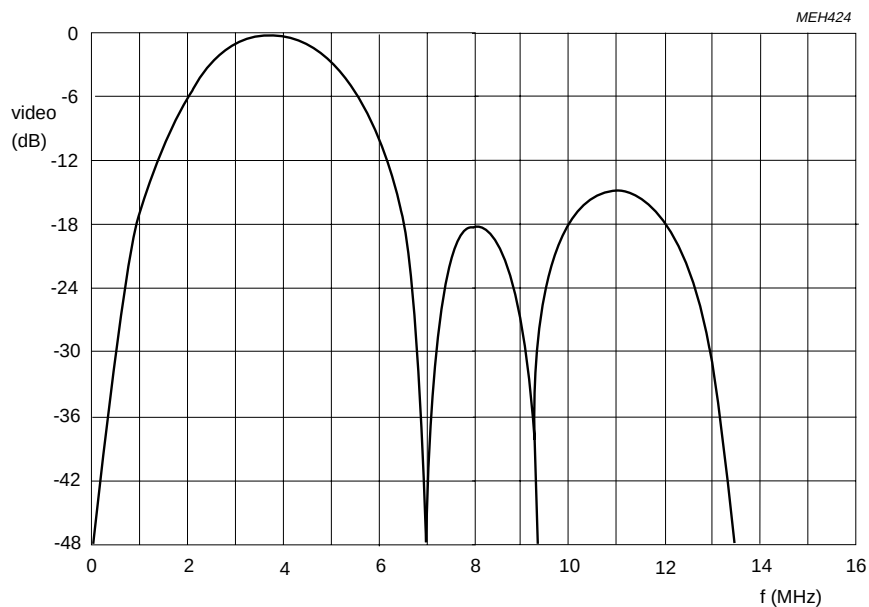


Fig.4 Frequency response of bandpass filters 1 and 2 with CFRQ-bit = 1.

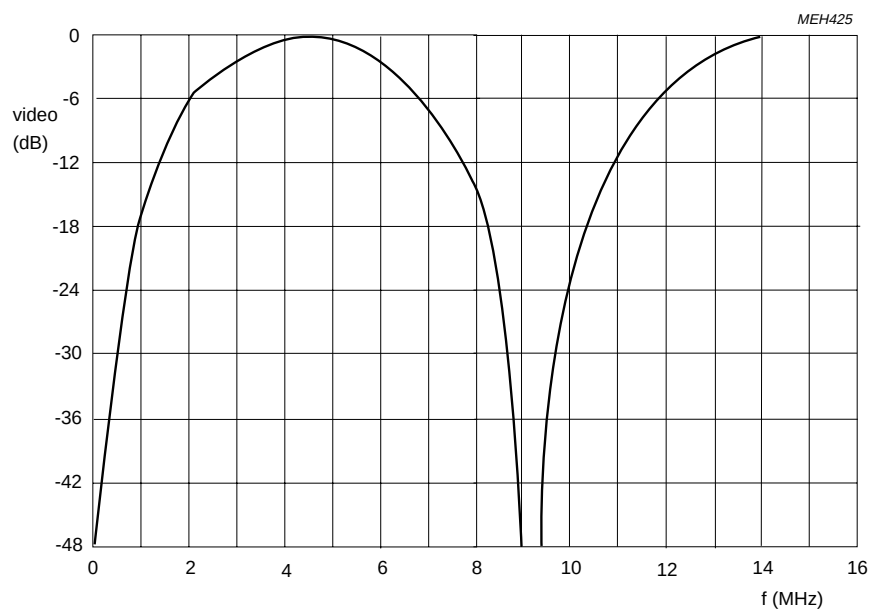


Fig.5 Frequency response of bandpass filters 1 and 2 with CFRQ-bit = 0.

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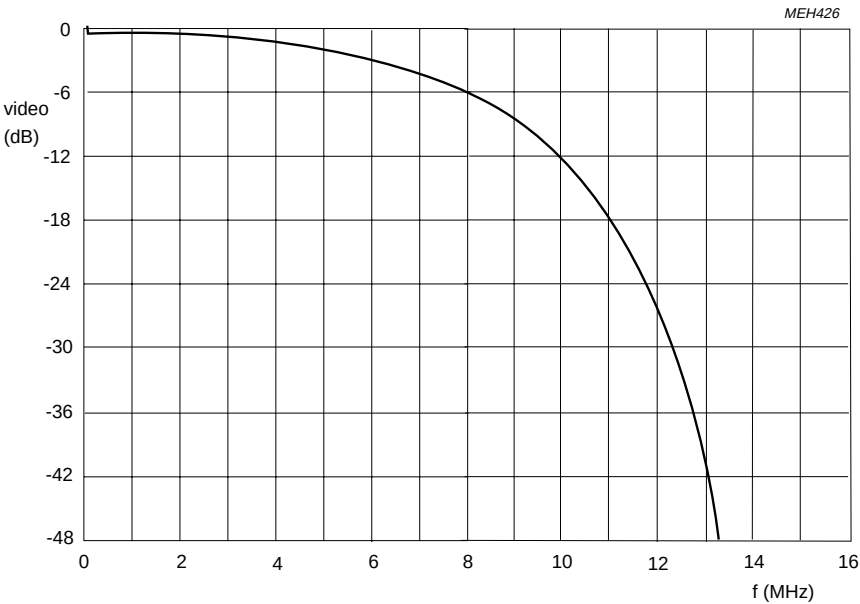


Fig.6 Frequency response of low-pass filter with CFRQ-bit = 1.

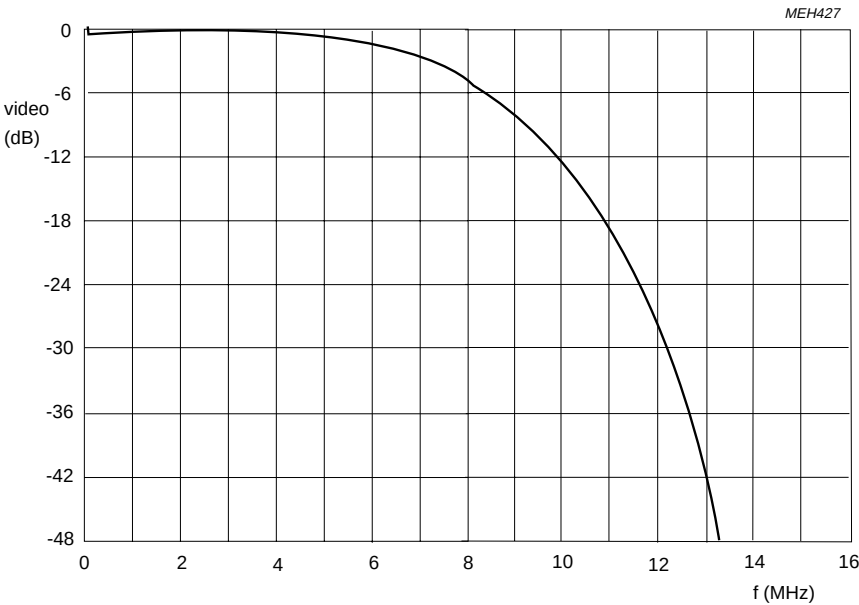


Fig.7 Frequency response of low-pass filter with CFRQ-bit = 0.

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## LIMITING VALUES

In accordance with the Absolute Maximum Rating System (IEC 134).

| SYMBOL    | PARAMETER                                          | MIN. | MAX.           | UNIT |
|-----------|----------------------------------------------------|------|----------------|------|
| $V_{DD}$  | supply voltage (pins 11, 34, 44)                   | -0.5 | 7.0            | V    |
| $V_I$     | voltage on all inputs                              | -0.5 | $V_{DD} + 0.5$ | V    |
| $V_O$     | voltage on all outputs ( $I_{O\ max} = 20\ mA$ )   | -0.5 | $V_{DD} + 0.5$ | V    |
| $P_{tot}$ | total power dissipation                            | —    | 1.0            | W    |
| $T_{stg}$ | storage temperature range                          | -65  | 150            | °C   |
| $T_{amb}$ | operating ambient temperature range                | 0    | 70             | °C   |
| $V_{ESD}$ | electrostatic handling <sup>(1)</sup> for all pins | —    | ±2000          | V    |

## Note

1. Equivalent to discharging a 100 pF capacitor through a 1.5 kΩ series resistor; inputs and outputs are protected against electrostatic discharge in normal handling. Normal precautions appropriate to handle MOS devices is recommended (see “Handling MOS Devices”).

## CHARACTERISTICS

$V_{DD1}$  to  $V_{DD3} = 5\ V$ ;  $T_{amb} = 0$  to  $70\ ^\circ C$  and measurements taken in Fig.1 unless otherwise specified.

| SYMBOL                                                        | PARAMETER                              | CONDITIONS                                          | MIN. | TYP. | MAX.         | UNIT |
|---------------------------------------------------------------|----------------------------------------|-----------------------------------------------------|------|------|--------------|------|
| $V_{DD}$                                                      | supply voltage range (pins 11, 34, 44) |                                                     | 4.5  | 5.0  | 5.5          | V    |
| $I_{DD}$                                                      | total supply current (pins 11, 34, 44) | $V_{DD} = 5\ V$ ; inputs LOW; outputs not connected | —    | 85   | 180          | mA   |
| <b>I<sup>2</sup>C-bus, SDA and SCL (pins 23 and 24)</b>       |                                        |                                                     |      |      |              |      |
| $V_{IL}$                                                      | input voltage LOW                      |                                                     | -0.5 | —    | 1.5          | V    |
| $V_{IH}$                                                      | input voltage HIGH                     |                                                     | 3    | —    | $V_{DD}+0.5$ | V    |
| $I_{23, 24}$                                                  | input current                          |                                                     | —    | —    | ±10          | μA   |
| $I_{ACK}$                                                     | output current on pin 23               | acknowledge                                         | 3    | —    | —            | mA   |
| $V_{OL}$                                                      | output voltage at acknowledge          | $I_{23} = 3\ mA$                                    | —    | —    | 0.4          | V    |
| <b>Data and clock inputs (pins 2 to 10 and pins 13 to 20)</b> |                                        |                                                     |      |      |              |      |
| $V_{IL}$                                                      | LL27 input voltage (pin 2)             | LOW                                                 | -0.5 | —    | 0.6          | V    |
| $V_{IH}$                                                      |                                        | HIGH                                                | 2.4  | —    | $V_{DD}+0.5$ | V    |
| $V_{IL}$                                                      | other input voltages                   | LOW                                                 | -0.5 | —    | 0.8          | V    |
| $V_{IH}$                                                      |                                        | HIGH                                                | 2.0  | —    | $V_{DD}+0.5$ | V    |
| $I_{leak}$                                                    | input leakage current                  |                                                     | —    | —    | 10           | μA   |
| $C_i$                                                         | input capacitance                      | data inputs                                         | —    | —    | 8            | pF   |
|                                                               |                                        | clock inputs                                        | —    | —    | 10           | pF   |
| $t_{SU,DAT}$                                                  | input data set-up time                 | Fig.8                                               | 11   | —    | —            | ns   |
| $t_{HD,DAT}$                                                  | input data hold time                   |                                                     | 3    | —    | —            | ns   |

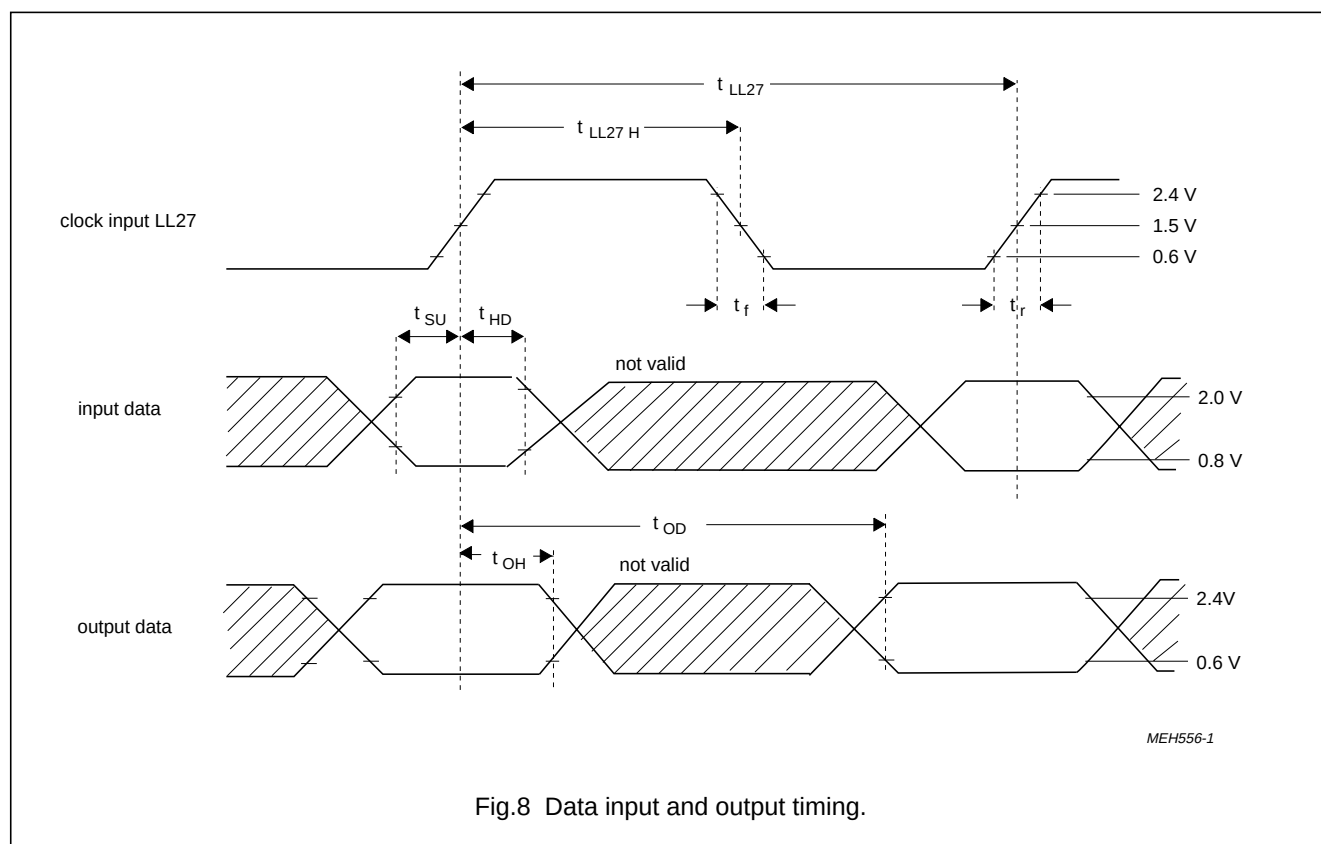
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| SYMBOL                                                | PARAMETER                                          | CONDITIONS             | MIN. | TYP. | MAX.     | UNIT |
|-------------------------------------------------------|----------------------------------------------------|------------------------|------|------|----------|------|
| <b>Data outputs (pins 25 to 32 and pins 35 to 42)</b> |                                                    |                        |      |      |          |      |
| $V_{OL}$                                              | output voltage LOW                                 |                        | 0    | –    | 0.6      | V    |
| $V_{OH}$                                              | output voltage HIGH                                |                        | 2.4  | –    | $V_{DD}$ | V    |
| $C_L$                                                 | load capacitor                                     |                        | 8    | –    | 25       | pF   |
| <b>Timing of data outputs</b>                         |                                                    | Fig.8                  |      |      |          |      |
| $t_{OH}$                                              | output signal hold time from positive edge of LL27 | $C_L = 8 \text{ pF}$   | 3    | –    | –        | ns   |
| $t_{OD}$                                              | output delay from positive edge of LL27            | $C_L = 25 \text{ pF}$  | –    | –    | 32       | ns   |
| <b>Line locked clock input LL27 (pin 2)</b>           |                                                    | Fig.8                  |      |      |          |      |
| $t_{LL27}$                                            | cycle time                                         | note 1                 | 35   | –    | 39       | ns   |
| $t_p$                                                 | duty factor                                        | $t_{LL27H} / t_{LL27}$ | 40   | 50   | 60       | %    |
| $t_r$                                                 | rise time                                          |                        | –    | –    | 5        | ns   |
| $t_f$                                                 | fall time                                          |                        | –    | –    | 6        | ns   |

**Note**

1.  $t_{SU}$ ,  $t_{HD}$ ,  $t_{OH}$  and  $t_{OD}$  include  $t_r$  and  $t_f$ .



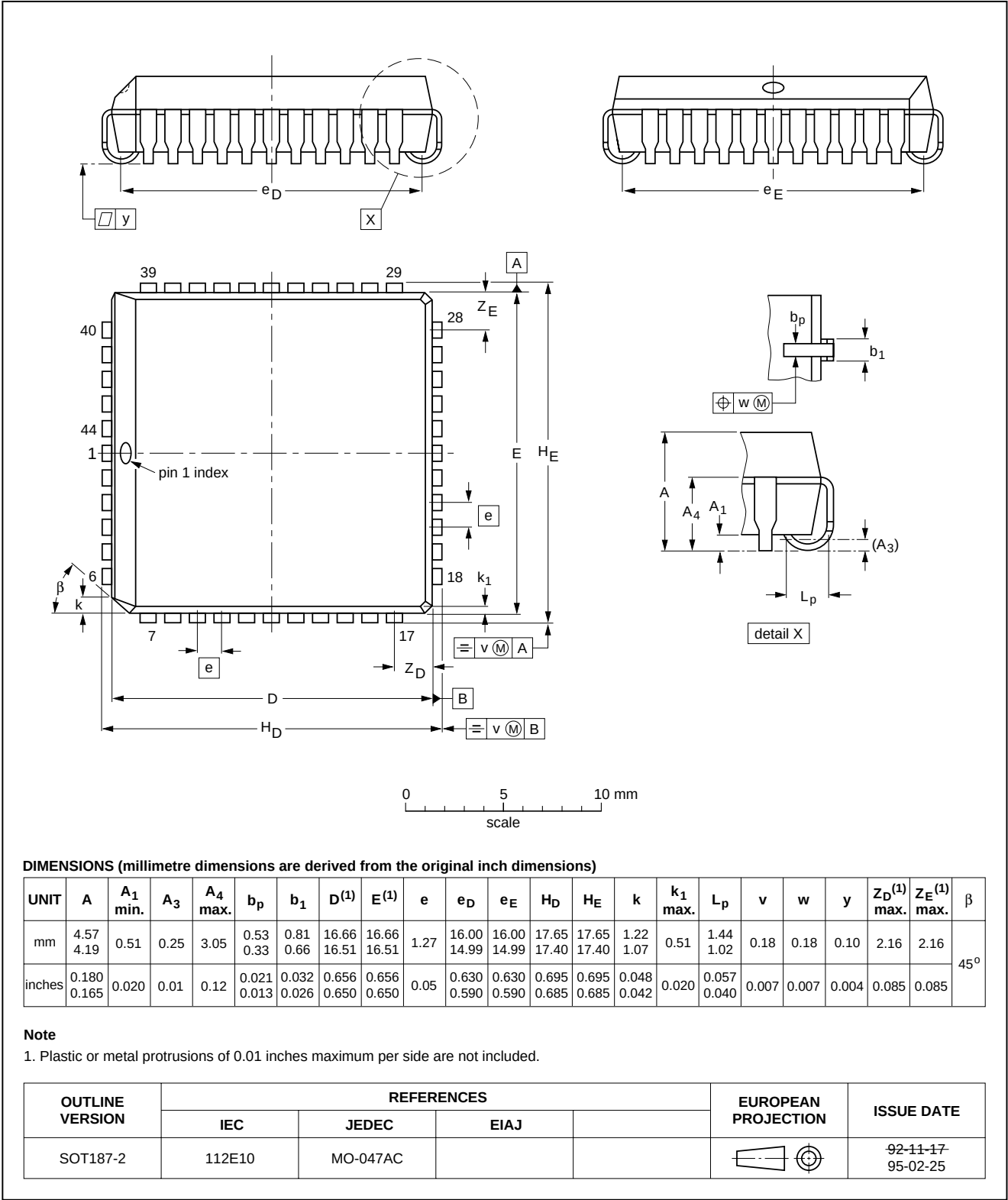
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PACKAGE OUTLINE

PLCC44: plastic leaded chip carrier; 44 leads

SOT187-2



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### SOLDERING

#### Introduction

There is no soldering method that is ideal for all IC packages. Wave soldering is often preferred when through-hole and surface mounted components are mixed on one printed-circuit board. However, wave soldering is not always suitable for surface mounted ICs, or for printed-circuits with high population densities. In these situations reflow soldering is often used.

This text gives a very brief insight to a complex technology. A more in-depth account of soldering ICs can be found in our *"IC Package Databook"* (order code 9398 652 90011).

#### Reflow soldering

Reflow soldering techniques are suitable for all PLCC and QFP packages.

The choice of heating method may be influenced by larger PLCC or QFP packages (44 leads, or more). If infrared or vapour phase heating is used and the large packages are not absolutely dry (less than 0.1% moisture content by weight), vaporization of the small amount of moisture in them can cause cracking of the plastic body. For more information, refer to the Drypack chapter in our *"Quality Reference Handbook"* (order code 9397 750 00192).

Reflow soldering requires solder paste (a suspension of fine solder particles, flux and binding agent) to be applied to the printed-circuit board by screen printing, stencilling or pressure-syringe dispensing before package placement.

Several techniques exist for reflowing; for example, thermal conduction by heated belt. Dwell times vary between 50 and 300 seconds depending on heating method. Typical reflow temperatures range from 215 to 250 °C.

Preheating is necessary to dry the paste and evaporate the binding agent. Preheating duration: 45 minutes at 45 °C.

#### Wave soldering

##### PLCC

Wave soldering techniques can be used for all PLCC packages if the following conditions are observed:

- A double-wave (a turbulent wave with high upward pressure followed by a smooth laminar wave) soldering technique should be used.
- The longitudinal axis of the package footprint must be parallel to the solder flow.

- The package footprint must incorporate solder thieves at the downstream corners.

##### QFP

Wave soldering is **not** recommended for QFP packages. This is because of the likelihood of solder bridging due to closely-spaced leads and the possibility of incomplete solder penetration in multi-lead devices.

**If wave soldering cannot be avoided, the following conditions must be observed:**

- A double-wave (a turbulent wave with high upward pressure followed by a smooth laminar wave) soldering technique should be used.
- The footprint must be at an angle of 45° to the board direction and must incorporate solder thieves downstream and at the side corners.

**Even with these conditions, do not consider wave soldering the following packages: QFP52 (SOT379-1), QFP100 (SOT317-1), QFP100 (SOT317-2), QFP100 (SOT382-1) or QFP160 (SOT322-1).**

##### METHOD (PLCC AND QFP)

During placement and before soldering, the package must be fixed with a droplet of adhesive. The adhesive can be applied by screen printing, pin transfer or syringe dispensing. The package can be soldered after the adhesive is cured.

Maximum permissible solder temperature is 260 °C, and maximum duration of package immersion in solder is 10 seconds, if cooled to less than 150 °C within 6 seconds. Typical dwell time is 4 seconds at 250 °C.

A mildly-activated flux will eliminate the need for removal of corrosive residues in most applications.

#### Repairing soldered joints

Fix the component by first soldering two diagonally-opposite end leads. Use only a low voltage soldering iron (less than 24 V) applied to the flat part of the lead. Contact time must be limited to 10 seconds at up to 300 °C. When using a dedicated tool, all other leads can be soldered in one operation within 2 to 5 seconds between 270 and 320 °C.

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## DEFINITIONS

| Data sheet status                                                                                                                                                                                                                                                                                                                                                                                                                                         |                                                                                       |
|-----------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|---------------------------------------------------------------------------------------|
| Objective specification                                                                                                                                                                                                                                                                                                                                                                                                                                   | This data sheet contains target or goal specifications for product development.       |
| Preliminary specification                                                                                                                                                                                                                                                                                                                                                                                                                                 | This data sheet contains preliminary data; supplementary data may be published later. |
| Product specification                                                                                                                                                                                                                                                                                                                                                                                                                                     | This data sheet contains final product specifications.                                |
| Limiting values                                                                                                                                                                                                                                                                                                                                                                                                                                           |                                                                                       |
| Limiting values given are in accordance with the Absolute Maximum Rating System (IEC 134). Stress above one or more of the limiting values may cause permanent damage to the device. These are stress ratings only and operation of the device at these or at any other conditions above those given in the Characteristics sections of the specification is not implied. Exposure to limiting values for extended periods may affect device reliability. |                                                                                       |
| Application information                                                                                                                                                                                                                                                                                                                                                                                                                                   |                                                                                       |
| Where application information is given, it is advisory and does not form part of the specification.                                                                                                                                                                                                                                                                                                                                                       |                                                                                       |

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